

How much of our current innovation work is guided by catchphrases that lack a scientific foundation?

A New Research Program

What comparable research programs could we be engaged in to discover how to work more effectively?

For one thing, we have very little understanding of what stimulates productivity under conditions of extreme uncertainty. Luckily, with cycle times falling everywhere, we have many opportunities to test new approaches. Thus, I propose that we create startup testing labs that could put all manner of product development methodologies to the test.

How might those tests be conducted? We could bring in small cross-functional teams, perhaps beginning with product and engineering, and have them work to solve problems by using different development methodologies. We could begin with problems with clear right answers, perhaps drawn from the many international programming competitions that have developed databases of well-defined problems with clear solutions. These competitions also provide a clear baseline of how long it should take for various problems to be solved so that we could establish clearly the individual problem-solving prowess of the experimental subjects.

Using this kind of setup for calibration, we could begin to vary the conditions of the experiments. The challenge will be to increase the level of uncertainty about what the right answer is while still being able to measure the quality of the outcome objectively. Perhaps we could use real-world customer problems and then have real consumers test the output of the teams' work. Or perhaps we could go so far as to build minimum viable products for solving the same set of problems over and over again to quantify which produces the best customer conversion rates.

We also could vary the all-important cycle time by choosing more or less complex development platforms and distribution channels to